

Shi Luode

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EDUCATION BACKGROUND

Shandong University (SDU)

09/2021-06/2025

- ♦ School of Information Science and Engineering (ISE)
- ♦ Major: Optical Information Science and Engineering | GPA: 86.09/100.0

RESEARCH & COMPETITION

Member, Infrared Remote Control Decoding

04/2024-06/2024

- ♦ Leveraged Altium Designer 10 to architect and draft circuit diagrams for decoding NEC and RC5 codes and designed PCB according to the diagrams.
- ♦ Authored initialization, counting value comparison, Timer 0 interrupt service routine, main infrared reception processing function, and digital tube display programs with C language.
- ♦ Resolved hardware connectivity and signal reception problems by interfacing with external pins, correcting digital tube orientation, reversing infrared receiver wiring, and incorporating a small lamp driver to simulate motor control, culminating in accurate digital tube display and remote-controlled lamp illumination.
- ♦ Optimized code logic for remote controls adhering to RC5 and NEC protocols, utilized guiding signals for automatic standard differentiation, and merged dual decoding codebases into a single script to ensure precise signal decoding.

Member, Intelligent Monitoring Security System Based on Machine Vision and Laser Guidance

10/2023-01/2024

- ♦ Collected and organized over 100 image datasets to build a new diverse dataset and utilized LabelImg to annotate the image data, ensuring the accuracy and completeness of the dataset.
- ♦ Trained the YOLOv5 model based on the PyTorch framework and adjusted hyperparameters to optimize model performance; conducted multi-object tracking experiments integrating the DeepSort algorithm, and assessed the identification and tracking accuracy with MOTA as an indicator of 29.1%.
- ♦ Chose MMTTracking framework as the base framework for target tracking, and made customized modifications to it in accordance with project requirements to meet the specific demands of the application scenario.

INTERN EXPERIENCE

Intern, Shandong Institutes of Industrial Technology

07/2024-08/2024

- ♦ Engaged in a project on defect detection for Hikvision robot model products (camera, lens, light source) during industrial production, testing the number of opened glass bottles in a frame on the production line according to customer requirements.
- ♦ Applied MXVision software to calibrate positions to ensure object recognition and manually adjusted the binary threshold to make continuously captured images of the opened bottle tops display higher RGB values.
- ♦ Utilized the Blob algorithm to filter out unnecessary counts (empty spaces, inverted bottles, unopened bottles) by area and aspect ratio.
- ♦ Processed the data using C# scripts and presented the results to the customer.

STUDENT WORK

Member, Student Department, SDU

04/2023-Present

- ♦ Participated in "Hope Guarding" Campaign of SDU, which was selected as part of the first batch of Innovation and Experimentation Projects for Serving Rural Revitalization at Universities Directly Under the Ministry of Education in 2021.
- ♦ Traveled to Henan province for Ice-breaking Operations and provided long-term online tutoring and mental health education for left-behind children.

Member, Management Department of Basketball Association, ISE

09/2022-09/2023

- ♦ Took charge of planning, staffing and selecting players for the ISE annual basketball tournament.
- ♦ Led the team to win the college basketball tournament championship for two consecutive years.

EXCHANGE EXPERIENCE

Shanghai Institute of Optics and Fine Mechanics Summer Camp, University of Chinese Academy of Sciences

05/2024

- ♦ Explored the "Laboratory of High Field Laser Physics" and the Station of Extreme Light and the Station of High Energy Density of Shanghai High repetition rate XFEL and Extreme light facility (SHINE).
- ♦ Engaged in expert lectures, participated in discussions, authored reports, and excelled in assessments, earning an Outstanding Camp Participant.

HONORS & AWARDS

Third-class Academic Scholarship, SDU

10/2023, 10/2024

ENGLISH & COMPUTER SKILLS

- ♦ English: IELTS: 6.5
- ♦ Computer: Proficient in C, C++, Python, MS Office, MATLAB, AutoCAD